

Khurana Lab 	SOP_VK_00xxx	Sequential extraction and immunoblotting	Author: Isabel (protocol from Volpicelli-Daley et al Nat Protocols 2014) Transcribed into Khurana Lab Protocol Format: Shawn Isabel	Revision:	issued
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1. Purpose

This protocol examines the fraction of alpha-synuclein (as assessed by alpha-synuclein and/or PS129 western blot) that is present in the triton-soluble or SDS-soluble fraction. Addition of alpha-synuclein pre-formed fibrils (PFFs) to neuron cultures seeds the recruitment of endogenous or transgenic alpha-synuclein into aggregates characterized by detergent insolubility.

Monomeric alpha-synuclein will be present in the triton-soluble fraction, whereas PFF-induced alpha-synuclein oligomers will be present in the triton-insoluble, SDS-soluble fraction.

2. Scope

This protocol encompasses preparation of cell extracts with Triton X-100, followed by sequential extraction of the Triton-insoluble material with SDS. The protein in the different detergent fractions are quantified by BCA, followed by SDS-PAGE and WB for alpha-synuclein, PS129, TUJ1, and loading control such as GAPDH.

3. Materials

50mL Falcon tube
 Bucket with ice
 Cell scraper
 Sonicator (I used Misonix Sonicator S-4000 from Wolfe Lab, next to ultracentrifuge)
 Ultracentrifuge (I used Optima MAX-TL Ultracentrifuge TLA-120.2 S/N 12U1811)
 polyallomar tube

4. Reagents

1X PBS
 1% (vol/vol) TX-100/TBS with protease and phosphatase inhibitors
 2% SDS/TBS with protease and phosphatase inhibitors
 4X Laemmli buffer

TBS/5% (wt/vol) milk
10X TBS
BCA assay reagents
SDS-PAGE reagents

5. Procedure

5.1 Rinse the neurons twice with PBS.

5.2 Place the dish on ice. By working one well at a time, completely aspirate the PBS and add the following volumes of ice-cold 1% (vol/vol) TX-100/TBS with protease and phosphatase inhibitors: 250 µl per well for a six-well plate, and 500 µl for a 6-cm dish.

5.3 Use a cell scraper to thoroughly scrape all neurons from each well.

5.4 Place the neurons in a polyallomer tube for a tabletop ultracentrifuge; keep the tube on ice.

5.5 Sonicate the tube ten times at a 0.5-s pulse and at 10% power.

- a. Used Misonix Sonicator S-4000, with Program Settings: Amplitude 10, Process Time 00:00:10, Pulse-ON time 00:00:01, Pulse-OFF time 00:00:01
- b. Wipe sonication tip with 1% SDS after coming into contact with PFFs, then 70% ethanol

5.6 Incubate the tube on ice for 30 min.

5.7 Centrifuge the tube at 100,000g at 4 °C for 30 min.

- a. Optima MAX-TL Ultracentrifuge TLA-120.2 S/N 12U1811 100,000xg 30 min 4C

5.8 Add 4× Laemmli buffer to ~150–200 µl of TX-100 supernatant (some of the supernatant is usually lost during Step 3–8; we recommend measuring the remaining supernatant before determining how much to remove for Laemmli buffer). Save ~20 µl of supernatant for protein assay. Retain the supernatant on ice or in a –20 °C freezer.

5.9 To the pellet, add the same volume of ice-cold 1% (vol/vol) TX-100/TBS with protease and phosphatase inhibitors: 250 µl per well for a six-well plate, and 500 µl for a 6-cm dish.

5.10 Sonicate ten times at a 0.5-s pulse and at 10% power. Keep the tip of the probe toward the bottom of the tube to prevent frothing. Make sure that the pellet is completely dispersed.

- a. Sonicator: Fisherbrand™ Model 120 Sonic Dismembrator (120V, 50/60Hz)
- b. I used Misonix Sonicator S-4000 with Program Settings: Amplitude 10, Process Time 00:00:10, Pulse-ON time: 00:00:01, Pulse-OFF time: 00:00:01

- 5.11 Centrifuge the mixture at 100,000g at 4 °C for 30 min.
- 5.12 Discard the supernatant.
- 5.13 Add 2% (vol/vol) SDS/TBS to the pellet with protease and phosphatases inhibitors. To a six-well plate, add 125 µl of 2% (wt/vol) SDS/TBS per well, and to a 6-cm dish add 250 µl of 2% (wt/vol) SDS/TBS.
- 5.14 Sonicate 15 times, at a 0.5-s pulse and at 10% power. Keep the tip of the probe toward the bottom of the tube. Make sure that the pellet is completely dispersed.
 - a. I used Misonix Sonicator S-4000 with Program Settings: Amplitude 10, Process Time 00:00:15, Pulse-ON time: 00:00:01, Pulse-OFF time: 00:00:01
- 5.15 Remove the supernatant and place it into a new microcentrifuge tube.
- 5.16 Perform a BCA/protein assay on TX-100 supernatant and SDS extract. Typically, a 1:5 dilution for the BCA assay is sufficient.
- 5.17 Dilute 2% (wt/vol) SDS extract from Step 15 into Laemmli buffer to 2× volume for the corresponding TX-100 fraction (regardless of the protein concentration of the SDS fraction). For example, if you have 180 µl of TX-100 extract (from Step 8) at 1 mg/ml and 90 µl of SDS extract, add 60 µl of 4× Laemmli buffer to the TX-100 extract and 30 µl of 4× Laemmli buffer to the SDS extract. Load 12.5 µl of both the TX-100 extract (10 g) and SDS extract. We suggest using a 2× volume because it makes the insoluble a-syn species more abundant and thus easier to visualize and quantify by immunoblotting.
- 5.18 Load the samples on a 4–20% (wt/vol) gel and run them according to the manufacturer's directions. We use 85 V constant voltage until the dye front runs off the gel. Be sure not to let the 10-kDa marker to run off the gel.
- 5.19 Transfer the proteins from the gel to a nitrocellulose membrane according to the manufacturer's instructions at 100 V for 75 min or overnight at 40 V.
- 5.20 Block the membrane for 1 h with TBS/5% (wt/vol) milk.
- 5.21 Dilute the primary antibodies in TBS/5% (wt/vol) milk and incubate them overnight at 4 °C with shaking.
- 5.22 Rinse the membrane three times with TBS/T, 10 min each rinse.
- 5.23 Incubate the membrane with HRP-conjugated secondary antibodies for 1 h at room temperature.
- 5.24 Rinse the membrane three times with TBS/T, 10 min each rinse.
- 5.25 Develop with enhanced chemiluminescence.

